

Customer Support Agent Evaluation Case Study

A sequential, evidence-first evaluation of a five-class routing agent

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Decision in one sentence

Retain the focused prompt as a development candidate, but do not release it on the current evidence: the frozen notebook improved from 91% to 98%, yet its 1.10% regression rate missed the preregistered 1% gate; the separate workbook-aligned candidate also remains Review because other precision is 0.882, below the 0.90 class-level bar.

Evidence at a glance

Evidence stream	Cases	Observed result	Decision supported
Supplied workbook audit	150	121 correct; 29 reviewed failures	Failure taxonomy and prompt hypothesis
Frozen notebook comparison	100	91 → 98; 8 wins; 1 regression	Focused V2 remains Review
Workbook-aligned controlled run	150	145 → 146; 1 win; 0 regressions	Separate candidate remains Review

Evidence boundary. These three streams use different source artifacts or label contracts. Their scores are reported independently and are never combined into a single improvement claim.

[GitHub repository](#) | [Google Doc](#) | [LangSmith comparison](#)

1. Product problem and evaluation objective

The system routes a customer-support ticket to one of five operational queues. A wrong route can delay resolution, transfer the customer between teams, or send an agent into a workflow without the tools needed to solve the request. The evaluation therefore treats classification quality as a product decision, not as a leaderboard score.

Routing contract

Category	Operational meaning	Boundary to protect
order_status	Tracking, current location, delivery ETA, missing-delivery investigation	Do not absorb general shipping questions or account-record problems
refund_request	Money back, credit, reversal, price adjustment, cancellation with returned funds	Resolve explicit remedy intent according to the dataset's label policy
product_issue	Received item is damaged, wrong, defective, or not as described	Separate post-purchase defects from pre-purchase questions
account_help	Login, password, saved address/payment method, checkout/account-access workflow	Route by the blocked account journey, not nearby product/order nouns
other	General information, feedback, stock, policy, browsing, sales, or gift-card questions	Avoid using other as a fallback for recognized operational failures

Evaluation question

Does a focused routing-policy change improve the highest-impact multi-intent failures while preserving previously correct behavior and keeping latency, token usage, and cost within acceptable bounds?

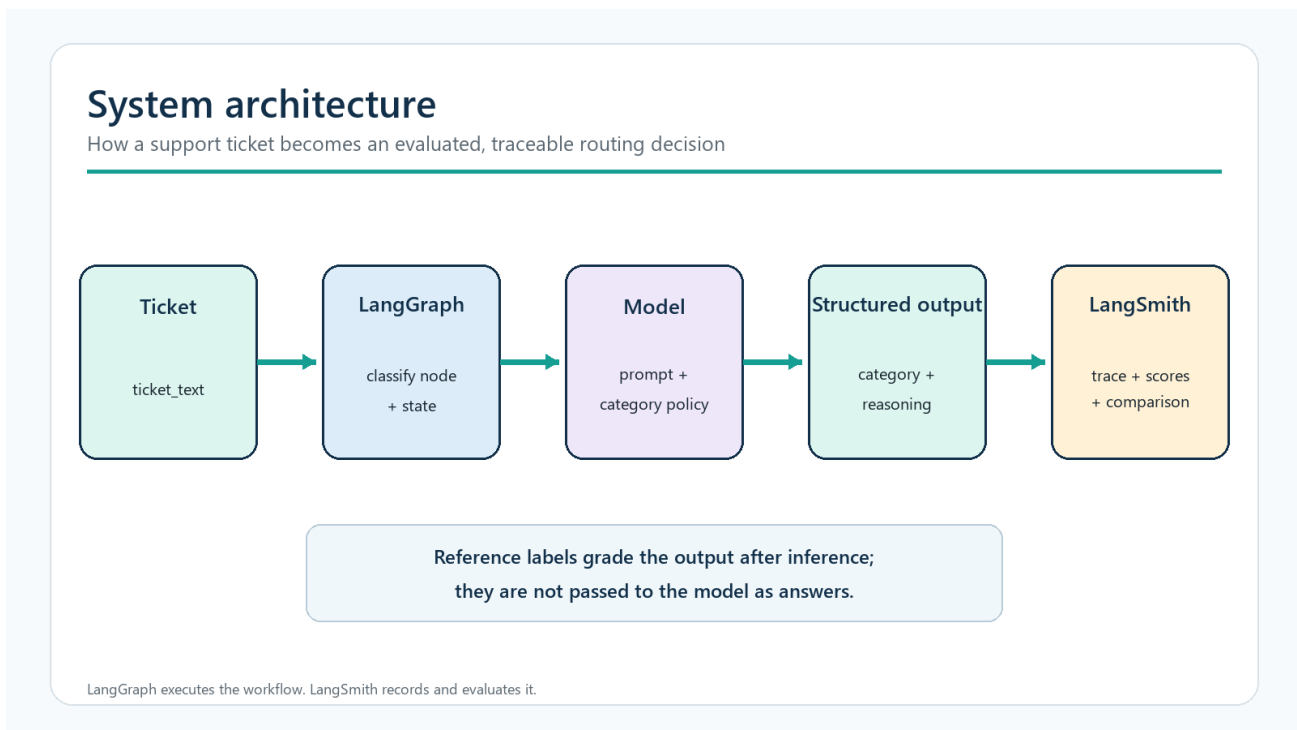


Figure 1. System under evaluation and the separation between inference inputs and reference labels.

2. Evaluation design

The workflow was designed around a repeatable product loop: define quality and risk, assemble and freeze evidence, run comparable experiments, analyze outcomes and traces, apply decision gates, then convert reviewed production failures into future regression cases.

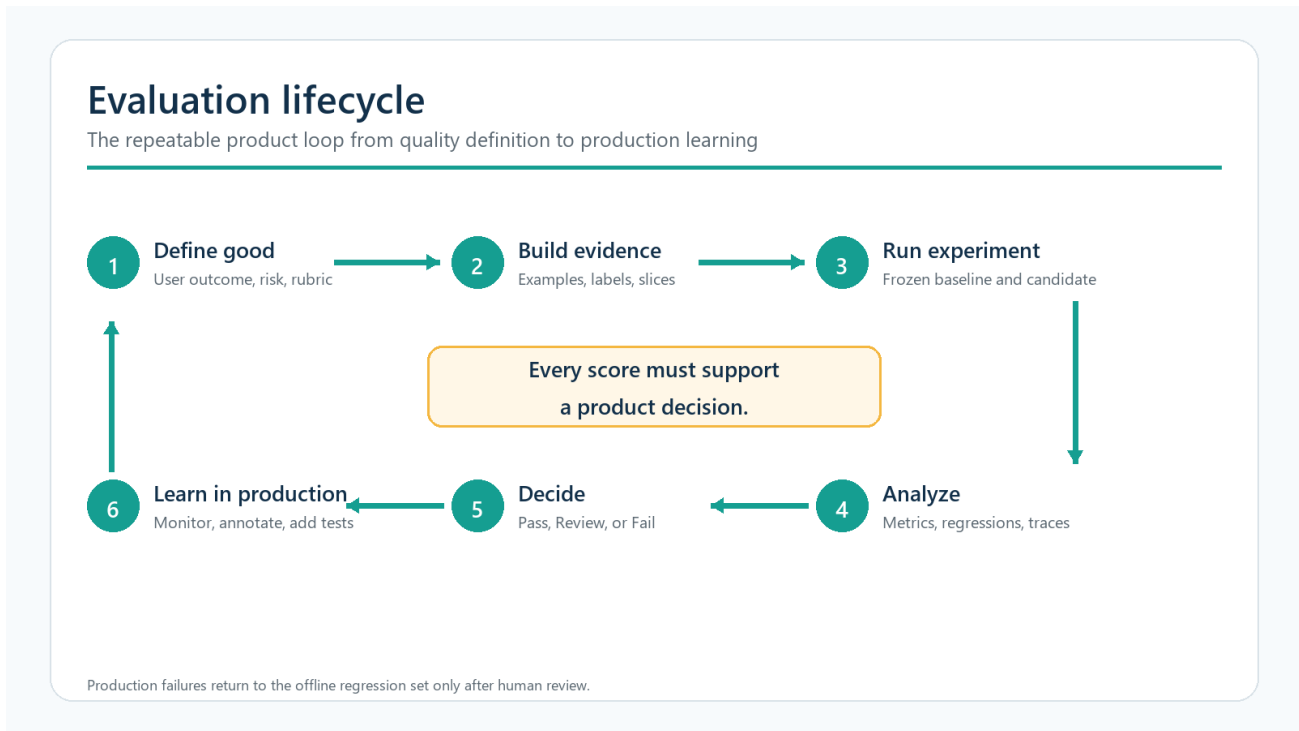


Figure 2. Evaluation lifecycle from product intent to monitored production learning.

Metrics and why they matter

Measure	Question answered	Decision use
Exact match	Did the predicted route equal the reference route?	Primary offline quality signal
Precision / recall / F1	Which queues are over- or under-routed?	Find class-specific risk hidden by aggregate accuracy
Allowed-category validity	Did the model produce one valid route?	Protect downstream automation and parsing
Paired outcomes	Which cases improved, regressed, stayed right, or stayed wrong?	Prevent aggregate gains from hiding regressions
Latency / tokens / cost	What operating price accompanies the quality change?	Assess customer experience and scale economics
Trace review	What input, reasoning, metadata, and component behavior caused a result?	Diagnose mechanisms and verify auditability

Release gates

Gate	Pass condition	Observed	Status
Allowed output	100% valid categories	100% in both notebook runs	PASS

Gate	Pass condition	Observed	Status
Net quality	Candidate improves exact match	91% → 98%	PASS
Regression budget	Right-to-wrong rate $\leq 1\%$ of V1-correct cases	1 / 91 = 1.10%	REVIEW
Class floor	Critical class precision ≥ 0.90	other precision = 0.882 in workbook-aligned V2	REVIEW
External validity	Independent holdout or production evidence	Not available	NOT ESTABLISHED

3. Evidence design and integrity controls

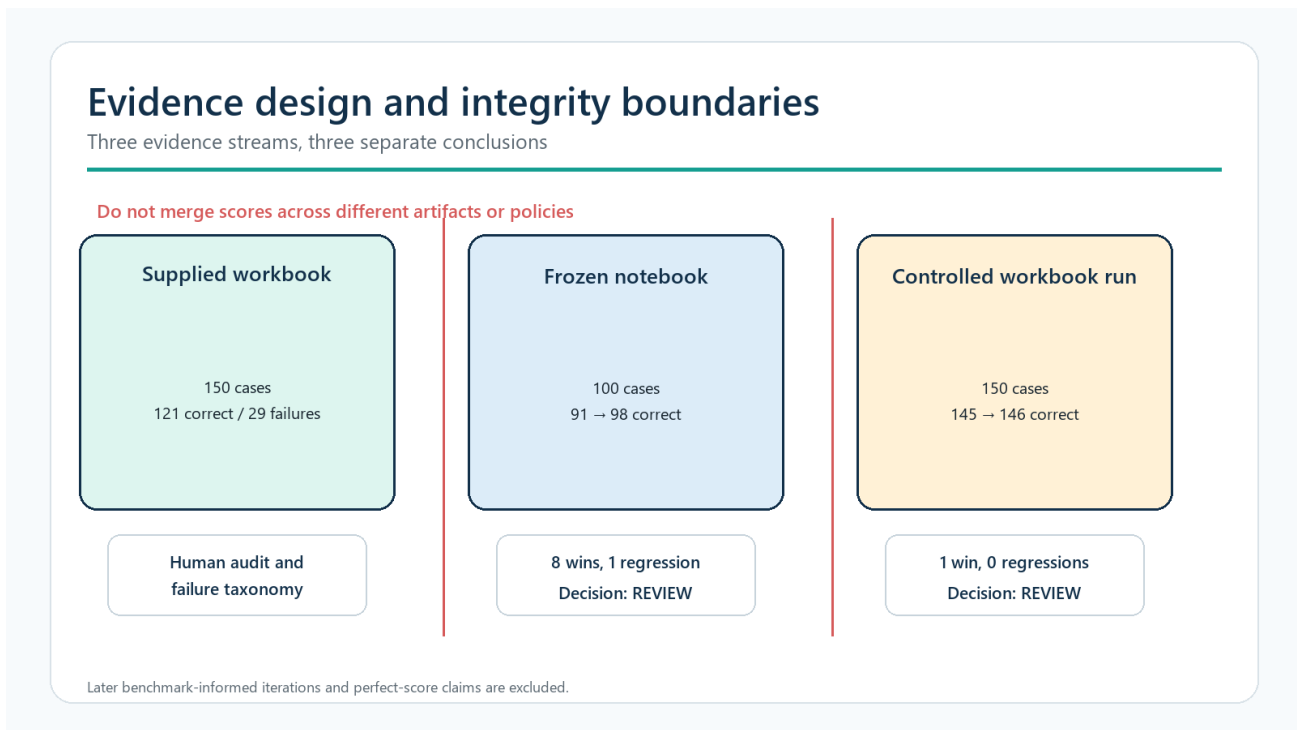


Figure 3. Three evidence streams and the conclusions each can support.

Why the streams stay separate

The supplied workbook audit evaluates historical predictions already present in the workbook. The frozen notebook comparison evaluates V1 and V2 on a separate 100-case generated dataset. The workbook-aligned controlled comparison runs two prompts on the fixed 150 workbook cases. Because the artifacts and label policies differ, 121/150 cannot be presented as the causal baseline for 146/150.

The key policy conflict

Ambiguous intent	Workbook policy	Notebook policy	Implication
Missing order + explicit refund request	refund_request	order_status	The same wording can have different correct labels
Damaged item + explicit refund request	refund_request for selected workbook cases	product_issue	A global refund-priority rule would damage notebook performance
Promo code rejected at checkout	account_help	May look like other under generic policy	Dataset-specific route ownership must be explicit

Interpretation. This conflict is not a modeling footnote. It is a product-policy decision. A production system needs one adjudicated routing contract before scores from different benchmarks can be compared or one prompt can be optimized across them.

Integrity controls used

- Stable case IDs and frozen V1/V2 exports for paired comparison.
- Reference labels used only by evaluators after inference, never passed to the classifier as answers.
- Prompt versions and SHA-256 hashes preserved with the result package.
- Every changed prediction reconciled into stable pass, win, regression, or remaining failure.
- Later benchmark-informed prompt versions, incomplete recovery runs, challenge-set experiments, and perfect-score claims excluded from the submission evidence.

4. Step 1 — Audit the supplied 150-row workbook

Every supplied prediction was reviewed against the ticket text and expected category. The workbook contained 121 passes and 29 failures, or 80.67% exact-match accuracy. The audit added a concrete human annotation to each failed row before grouping the failures.

Reference class	Precision	Recall	F1	Support
Order status	0.629	0.917	0.746	24
Refund request	1.000	0.714	0.833	42
Product issue	0.852	0.958	0.902	24
Account help	0.808	0.700	0.750	30
Other	0.781	0.833	0.806	30
Macro average	0.814	0.825	0.808	150

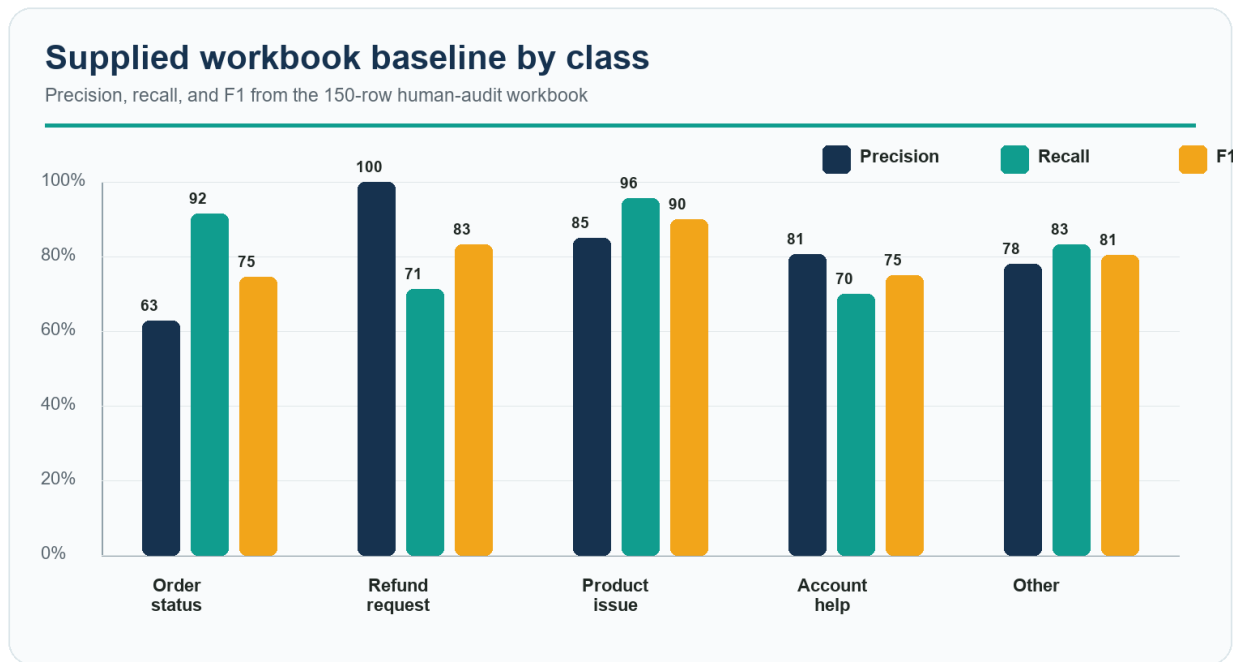


Figure 4. Supplied workbook baseline metrics by class.

What the class metrics reveal

- refund_request precision is 1.000 but recall is only 0.714: when the model predicts refund it is usually right, but it misses many true refund cases.
- order_status recall is high at 0.917 while precision is only 0.629, consistent with missing-order refund requests being absorbed into the status queue.
- account_help recall is 0.700, indicating that checkout, account-history, and promo-code failures are often routed by nearby nouns instead of the blocked workflow.
- Macro F1 is 0.808, confirming that the aggregate 80.67% accuracy is not hiding a single dominant high-performing class.

5. Steps 2 and 3 — Convert errors into a failure taxonomy

The 29 annotations were clustered by decision mechanism rather than by surface wording. Every failed row was assigned to one of four mutually usable categories so the prompt hypothesis could target a policy boundary instead of memorizing examples.

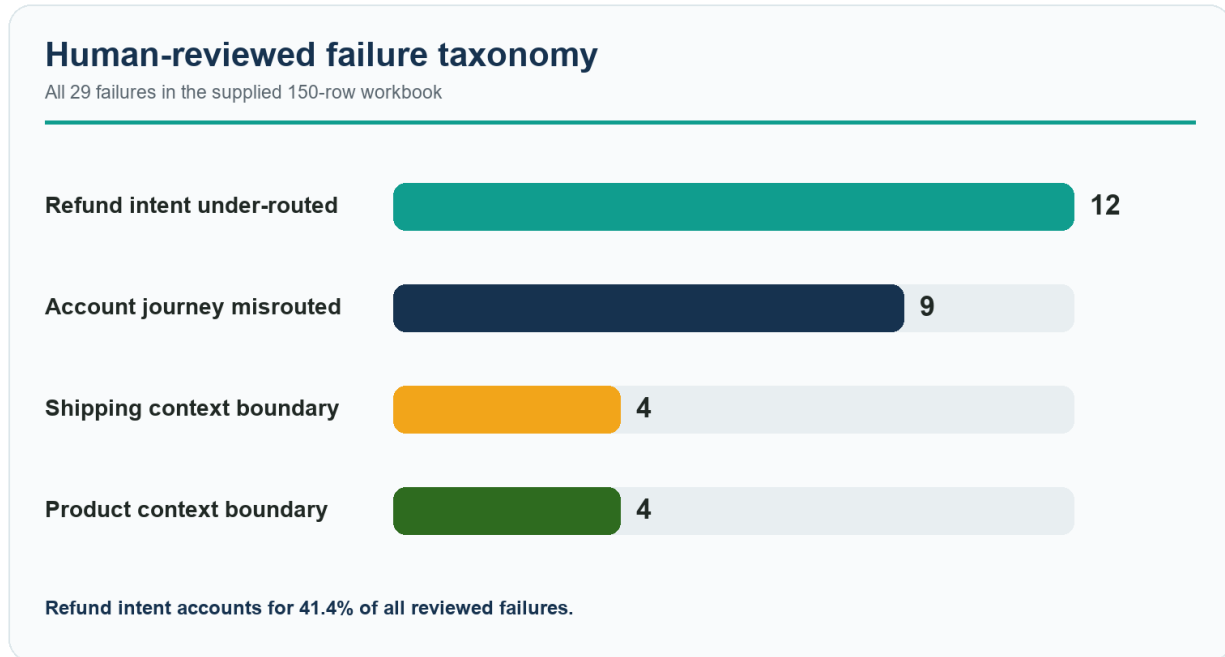


Figure 5. Distribution of the 29 human-reviewed workbook failures.

Failure group	Count	Definition	Product impact
Refund Intent Underouted	12	Explicit refund, money-back, price-adjustment, or cancellation-plus-refund intent routed elsewhere	Delays monetary resolution; adds transfers and escalation risk
Account Journey Misrouted	9	Checkout, account visibility, or session failures routed by nearby product/order words	Leaves customers blocked from payment or purchase records
Shipping Context Boundary	4	Delivery incidents confused with profile changes or general shipping feedback	Sends cases to teams with the wrong tools and SLA
Product Context Boundary	4	Post-purchase claim failure or pre-purchase question assigned to the wrong side of the boundary	Creates avoidable return work or leaves claims unresolved

Representative annotated failures

ID	Ticket	Reference	Prediction	Human diagnosis
t012	I never received my order and I want my money back.	refund_request	order_status	Model followed missing-order context and ignored the requested monetary outcome.
t024	The carrier marked it delivered but it shows the wrong delivery address.	order_status	account_help	Model mapped 'address' to profile support although this is a delivery incident.
t103	Site logs me out every time I try to pay.	account_help	product_issue	Model treated the website malfunction as a product defect instead of a blocked checkout session.
t144	Are the AirPods you sell authentic Apple products?	other	product_issue	Model treated a pre-purchase authenticity question as a received-item defect.

6. Step 4 — Form a focused prompt hypothesis

Refund Intent Underouted was selected because it was the largest group at 12 of 29 failures and because it directly affects money movement, trust, and escalation. The workbook prompt revision added one rule: explicit requests for a refund, money back, post-purchase price adjustment, or cancellation with returned funds should route to refund_request even when secondary context is present.

Design element	Baseline behavior	Focused change	Declared risk
Decision signal	Keyword/topic proximity	Use the customer's requested outcome	Overweighting one outcome can damage other policy contracts
Refund boundary	Refund language could lose to shipping/account context	Prioritize explicit monetary remedy under the workbook policy	Status or account cases containing refund language may over-route
Non-refund routes	Broad definitions without precedence	Keep status, replacement/repair, and account access when no monetary remedy is requested	Ambiguous compound requests still require policy adjudication
Prompt form	Five label definitions	One focused boundary rule; no case IDs or answer list	General rule may still encode benchmark-specific policy

Pre-run hypothesis

- Expected improvement: correct missing-order, price-adjustment, and subscription-cancellation cases labeled refund_request in the workbook.
- Regression risk: cases that mention refund language but are labeled by delivery or product context may move in the wrong direction.
- Evaluation safeguard: rerun on the same frozen IDs and inspect every changed prediction, not only the average score.

Important constraint. The notebook's 100-case label policy prioritizes underlying order or product context for several identical multi-intent patterns. The notebook V2 prompt therefore operationalizes its own frozen label contract. The two V2 results share a version name but are not one universal policy.

7. Run the frozen notebook V1 and V2 comparison

Experiment property	Value
Dataset	100 frozen development examples; 20 references per class
Case identity	Stable IDs t000–t099 preserved across V1 and V2
Model	gpt-4o-mini as recorded in LangSmith run metadata
Output schema	Structured category, reasoning, and ticket text
Evaluators	Allowed-category validity and deterministic category exact match
Tracing	LangSmith root runs with dataset version, example ID, prompt version, model, runtime, latency, tokens, and cost

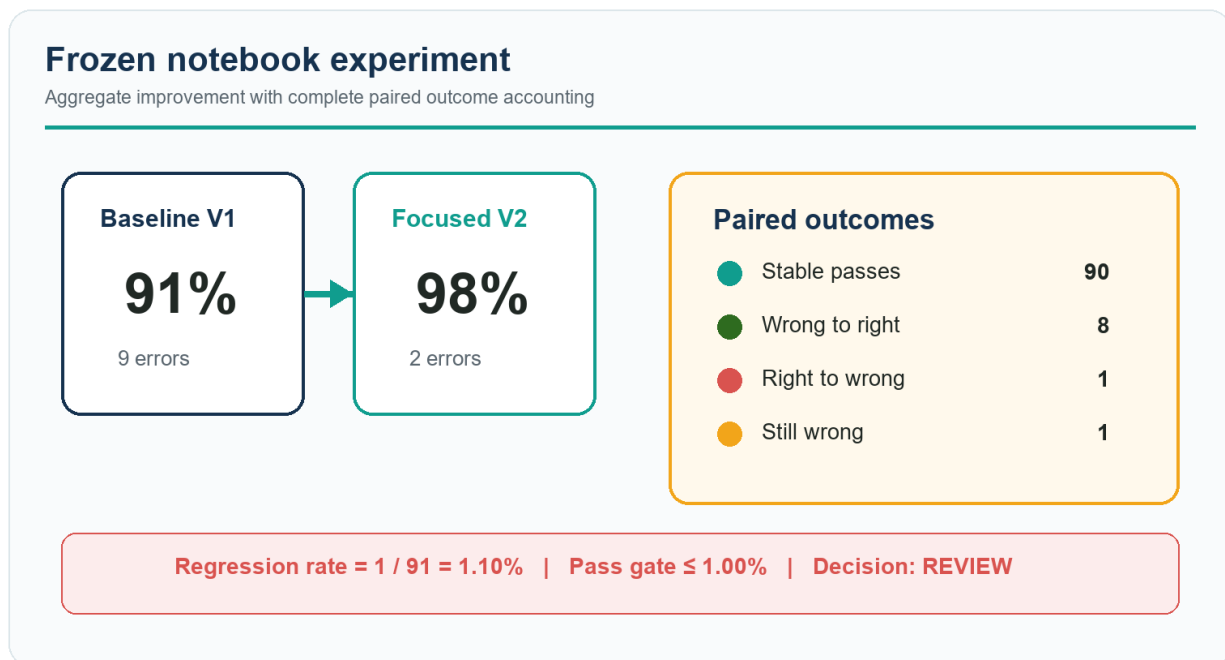


Figure 6. Frozen notebook aggregate results and paired outcome accounting.

Aggregate result

Baseline V1 scored 91/100 and focused V2 scored 98/100, a seven-percentage-point increase and a 77.8% relative reduction in observed errors. Both experiments returned an allowed category for all 100 examples.

Outcome	Cases	Meaning
Stable pass	90	Correct in both versions
Wrong to right	8	Candidate corrected a baseline failure
Right to wrong	1	Candidate introduced a regression
Still wrong	1	Neither version matched the reference

8. Inspect class-level movement

Class	V1 F1	V2 F1	Δ F1	Interpretation
Order status	0.889	1.000	+0.111	Four missing-order cases corrected
Refund request	0.833	1.000	+0.167	Precision recovered from over-routing
Product issue	0.857	0.947	+0.090	Improved, but two misleading-description cases remain wrong
Account help	1.000	1.000	+0.000	Stable perfect score on this set
Other	0.976	0.952	-0.023	Recall stayed perfect; precision declined because two product cases moved here

Frozen notebook class-level F1

Aggregate accuracy improved, while the product/other boundary remained imperfect

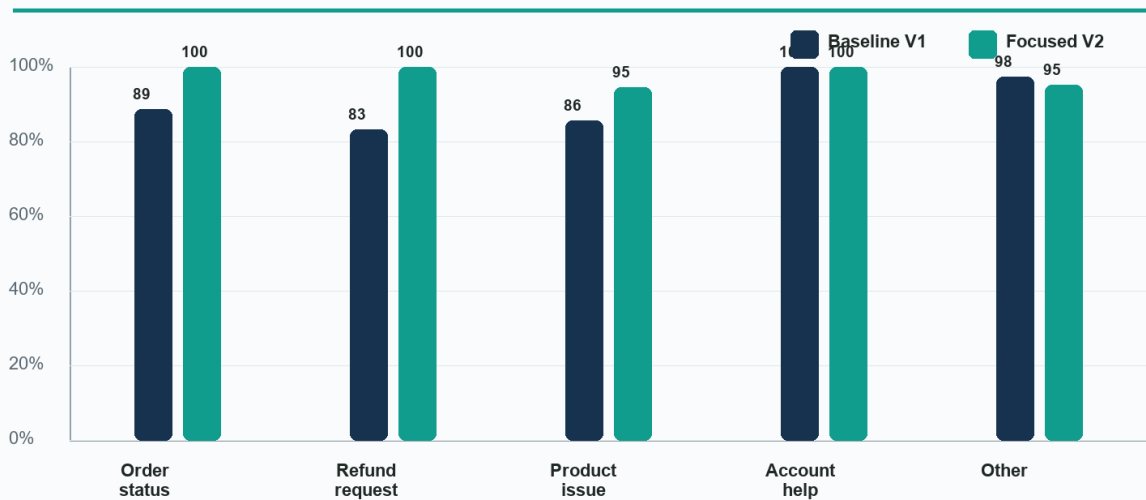


Figure 7. Per-class F1 reveals the remaining product/other boundary risk.

Confusion matrices

Rows are reference labels; columns are predicted labels

Baseline V1 — 91/100

	order	refund	product	account	other
order	16	4	0	0	0
refund	0	20	0	0	0
product	0	4	15	0	1
account	0	0	0	20	0
other	0	0	0	0	20

Focused V2 — 98/100

	order	refund	product	account	other
order	20	0	0	0	0
refund	0	20	0	0	0
product	0	0	18	0	2
account	0	0	0	20	0
other	0	0	0	0	20

Figure 8. Confusion matrices before and after the notebook prompt revision.

Reading the matrix. V1 over-routed four order-status and four product-issue references to refund_request, plus one product case to other. V2 repaired the refund over-routing, leaving two product_issue references predicted as other.

9. Inspect every changed prediction

ID	Ticket	Reference	V1	V2	Outcome
t004	I never received my order and I want my money back.	Order status	Refund request	Order status	WIN
t005	I've been waiting for my order, and it's still not here! I demand a refund immedi...	Order status	Refund request	Order status	WIN
t006	Hello, I'm reaching out because my order hasn't arrived yet. Could you please he...	Order status	Refund request	Order status	WIN
t007	Order #98765 is missing, and I'm pretty upset about it. Can you just refund me a...	Order status	Refund request	Order status	WIN
t044	My laptop arrived damaged and I want a full refund, not a replacement.	Product issue	Refund request	Product issue	WIN
t045	I received my laptop in a damaged condition and I expect a complete refund, not ...	Product issue	Refund request	Product issue	WIN
t046	I'm really unhappy because my laptop showed up broken, and I just want my money ...	Product issue	Refund request	Product issue	WIN
t047	The laptop I ordered came in with damage and I'd prefer a full refund rather tha...	Product issue	Refund request	Product issue	WIN
t052	The product description said 'wireless' but I had to buy a separate dongle to us...	Product issue	Product issue	Other	REGRESSION

What changed mechanistically

The eight wins are concentrated in two four-case families. Missing-order tickets labeled `order_status` stopped being captured by refund language. Damaged-product tickets labeled `product_issue` likewise stopped being captured by a requested refund. This concentration supports the intended precedence-rule mechanism on the notebook label policy.

The one regression is t052. The reference is `product_issue` because the purchased product failed an advertised wireless claim; V2 chose other because it interpreted the message as general product-information feedback. A paraphrase, t054, remained wrong for the same reason. The remaining risk is therefore a coherent product-description boundary, not random noise.

Personal / Datasets & Experiments / customer-support-routing-eval-... / Comparing 2 Experiments

Comparing cs-routing-baseline-v1-1... cs-routing-focused-v2-31... [+ Experiment](#)

Charts

Download, Filter, Sort, Columns, Default, Table Icons

	Inputs	Reference Outputs	Outputs	category_exact_match	
			A B	A B	
				0.91 AVG 0.98 AVG	
1	Order #98765 is missing, and I'm pretty ... set about it. Can you just refund me ahead	order_status	refund_request	order_status	0.00 1.00
2	The laptop I ordered came in with dama... and I'd prefer a full refund rather than a re	product_issue	refund_request	product_issue	0.00 1.00
3	My laptop arrived damaged and I want a ... ull refund, not a replacement.	product_issue	refund_request	product_issue	0.00 1.00
4	Hello, I'm reaching out because my orde... hasn't arrived yet. Could you please help	order_status	refund_request	order_status	0.00 1.00
5	I never received my order and I want my ... oney back.	order_status	refund_request	order_status	0.00 1.00
6	I'm really unhappy because my laptop sh... wed up broken, and I just want my money	product_issue	refund_request	product_issue	0.00 1.00
7	I've been waiting for my order, and it's st... not here! I demand a refund immediately!	order_status	refund_request	order_status	0.00 1.00
8	I received my laptop in a damaged condi... on and I expect a complete refund, not a r	product_issue	refund_request	product_issue	0.00 1.00

Figure 9. LangSmith comparison filtered to the eight wrong-to-right changes.

Personal / Datasets & Experiments / customer-support-routing-eval-... / Comparing 2 Experiments

Comparing cs-routing-baseline-v1-1... cs-routing-focused-v2-31... [+ Experiment](#)

Charts

Download, Filter, Sort, Columns, Default, Table Icons

	Inputs	Reference Outputs	Outputs	category_exact_match	1 regression
			A B	A B	
				0.91 AVG 0.98 AVG	
1	The product description said 'wireless' b... I had to buy a separate dongle to use it. Mi	product_issue	product_issue other	1.00 0.00	

Figure 10. LangSmith comparison filtered to the single right-to-wrong regression.

10. Connect aggregate scores to trace evidence

Aggregate metrics show what changed; traces explain how the system behaved on a concrete request. The LangSmith runs preserve the model input, structured output, reasoning, evaluation scores, latency, token usage, cost, runtime, dataset version, example ID, and prompt version.

Representative corrected trace

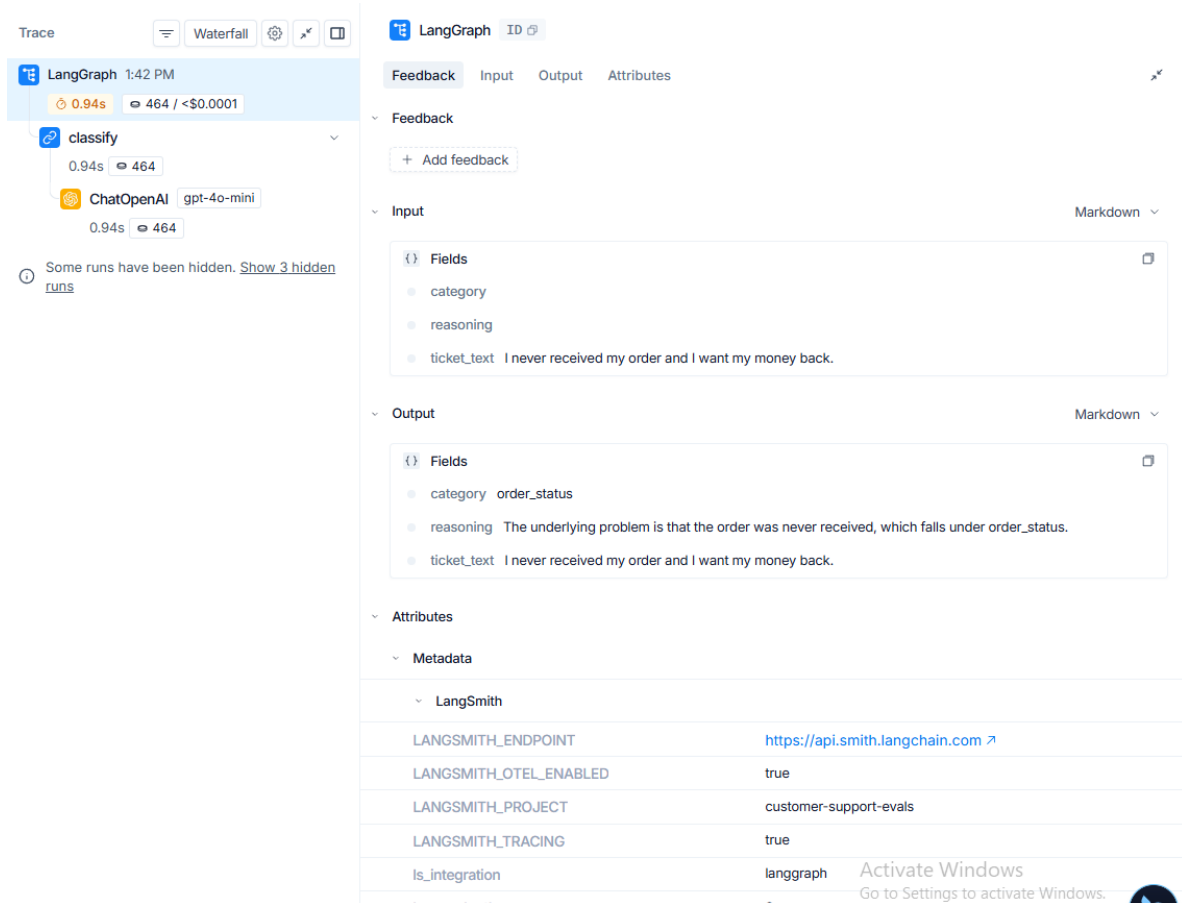


Figure 11. Corrected order-status trace with input, output, reasoning, and run metadata.

The corrected missing-order ticket now routes to `order_status` under the notebook policy. The trace makes the decision auditable and ties the row-level outcome to the aggregate eight-win count.

Representative regression trace

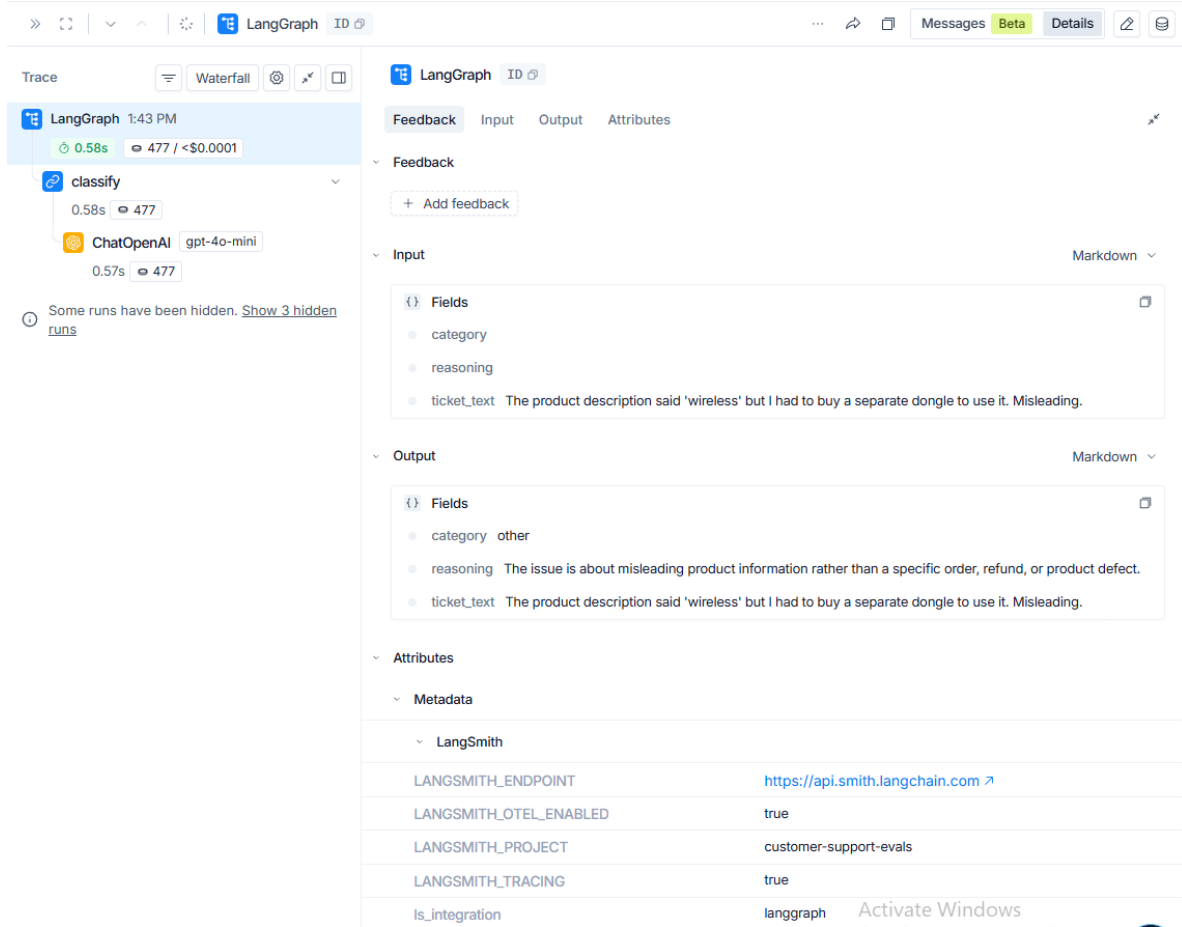


Figure 12. Regression trace for the misleading wireless-product-description ticket.

The regression trace preserves the counterexample needed for the next evaluation cycle. It should become a high-risk product-description slice in any future independently authored validation set.

11. Evaluate the operational trade-off

Measure	Baseline V1	Focused V2	Change
Exact match	91.0%	98.0%	+7.0 pp
Allowed-category validity	100.0%	100.0%	No change
Mean latency	0.773 s	0.855 s	+0.082 s
P95 latency	1.028 s	1.510 s	+0.481 s
P99 latency	1.990 s	3.625 s	+1.635 s
Mean tokens	241.52	476.46	+234.94
Total tokens	24,152	47,646	+23,494
Total cost / 100	\$0.0046884	\$0.0082998	+\$0.0036114

Operational trade-off

Focused V2 improved quality while increasing latency, tokens, and cost

Measure	Baseline V1	Focused V2	Relative change
Mean latency	0.773 s	0.855 s	+10.6%
P95 latency	1.028 s	1.510 s	+46.8%
Mean tokens	241.52	476.46	+97.3%
Total cost / 100	\$0.00469	\$0.00830	+77.0%

Allowed-category validity remained 100% in both runs.

Figure 13. Quality improved with a measurable latency, token, and cost increase.

Product interpretation. The absolute cost increase is tiny for 100 runs, but relative cost rises about 77% and mean tokens nearly double. At production scale, the decision should be based on avoided misroutes per incremental dollar and the customer impact of higher tail latency. The current data supports keeping V2 as a candidate, not deploying it without additional validation.

12. Apply the release decision

Why 98% still means Review

Aggregate improvement does not override a preregistered regression gate

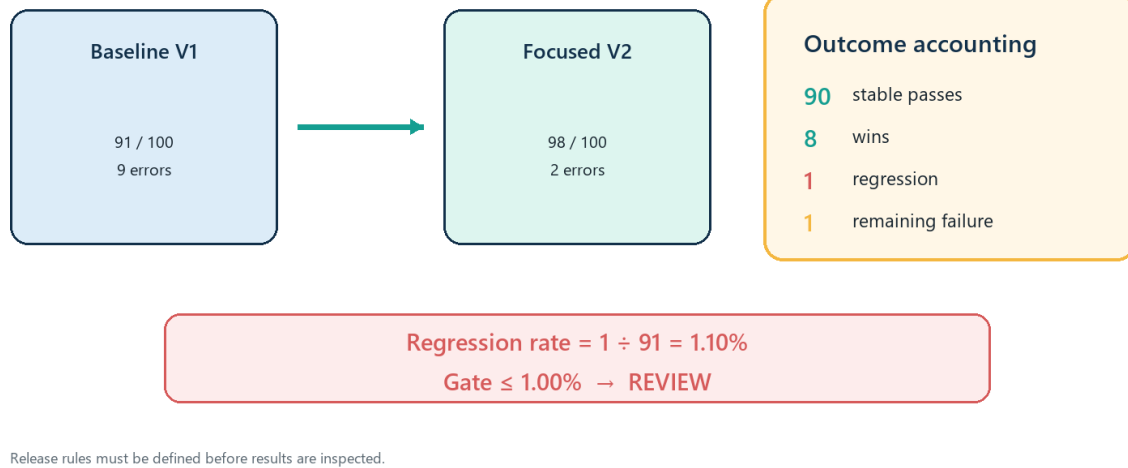


Figure 14. The regression gate converts the 98% aggregate score into a Review decision.

Decision criterion	Evidence	Assessment
Meaningful quality gain	+7 percentage points; eight wins; error count 9 → 2	Meets
No invalid outputs	100/100 allowed categories in both runs	Meets
Regression rate within budget	1 regression / 91 previously correct = 1.10%; gate $\leq 1\%$	Misses narrowly
Remaining failure understood	t052 regression and t054 remaining failure share one product-description boundary	Understood; unresolved
Operational cost acceptable	+10.6% mean latency; +46.8% p95; +97.3% mean tokens; +77.0% total cost	Requires product trade-off
Independent generalization	No organizer-hidden or independently authored holdout result	Not established

Decision — Review

Retain V2 as a development candidate because it materially improves the frozen notebook set and its remaining errors are interpretable. Do not release solely on this evidence. The candidate misses the explicit regression gate, increases operating cost, and has not been evaluated on an independently authored holdout.

13. Separate workbook-aligned controlled comparison

A second controlled experiment ran the original and focused workbook prompts on the same fixed 150 workbook cases. Controlled V1 scored 145/150 and focused V2 scored 146/150. This experiment is reported separately from the supplied workbook's historical 121/150 predictions because 25 supplied predictions changed when the original prompt was rerun under the current model/runtime.

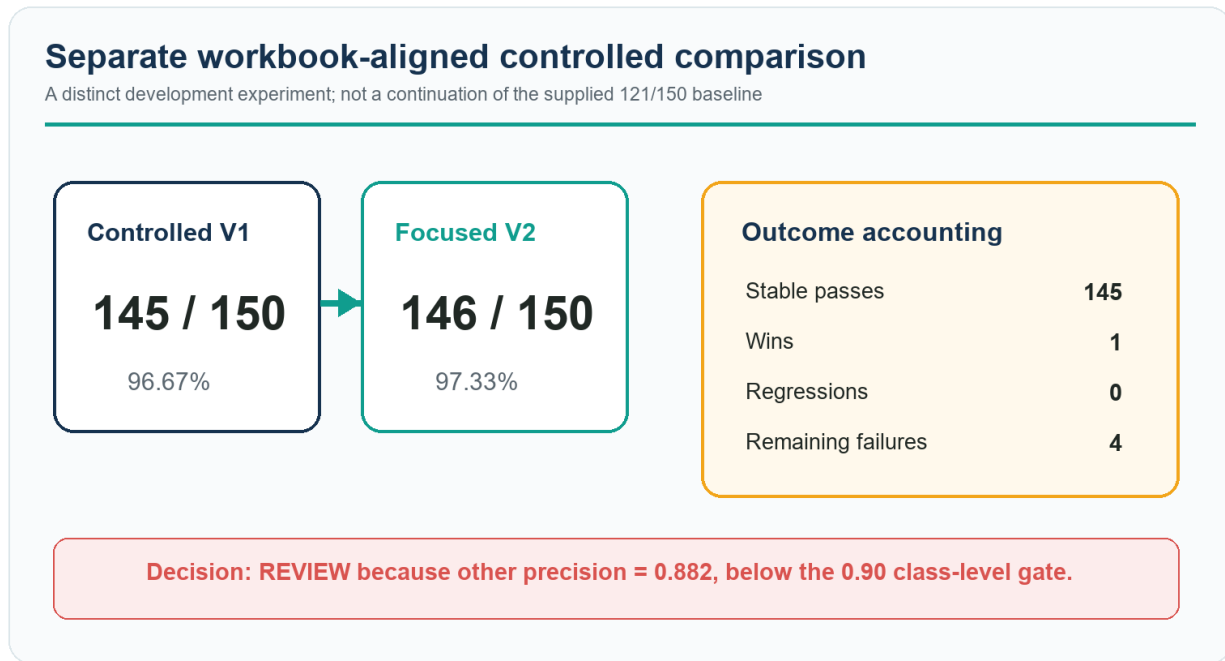


Figure 15. Workbook-aligned controlled V1/V2 comparison and decision.

Measure	Controlled V1	Focused V2	Change
Exact match	145 / 150 (96.67%)	146 / 150 (97.33%)	+1 case / +0.67 pp
Macro F1	0.966	0.975	+0.008
Wins / regressions	—	1 / 0	Positive paired result
Mean latency	0.740 s	0.771 s	+0.031 s
Mean tokens	231.49	298.14	+66.65
Total cost	\$0.006843	\$0.008387	+\$0.001543

What actually improved

The only wrong-to-right change was t025: a carrier delivered a parcel to the wrong address. Controlled V1 predicted product_issue; focused V2 predicted order_status, matching the reference. The intended refund cluster already had little headroom in controlled V1, so this experiment does not prove a large refund-policy effect.

Four remaining failures

ID	Reference	V2 prediction	Failure mechanism
t050	Refund request	Other	Refund Intent Underouted
t114	Account help	Other	Account Journey Misrouted
t115	Account help	Other	Account Journey Misrouted
t116	Account help	Other	Account Journey Misrouted

Decision. Focused V2 remains Review because other precision is 0.882, below the 0.90 class-level bar. The controlled comparison is useful reproducibility and robustness evidence, but its +1 result is modest and should not be described as a direct improvement from 121/150 to 146/150.

14. What the evaluation establishes — and what it does not

Claim	Evidence status	Reason
The supplied model has recurring policy-boundary failures	Established	All 29 supplied failures were manually annotated and grouped
Notebook V2 improves the frozen 100-case development set	Established	Same IDs and references; 8 wins, 1 regression, 1 remaining failure
Workbook-focused V2 modestly improves its controlled V1	Established	Same 150 cases; 1 win and 0 regressions
121/150 causally becomes 146/150	Not supported	The 121 score is a different historical prediction artifact
The prompt generalizes to unseen organizer or production data	Not established	No independent hidden or production set is available
The optional LLM judge is calibrated and production-ready	Not claimed	The workbook contains scaffolding, but the judge extension was not completed as submission evidence

Threats to validity

Threat	Observed condition	Mitigation / next test
Label-contract drift	Workbook and notebook disagree on multi-intent precedence	Adjudicate one production routing policy before cross-set comparison
Development-set dependence	Prompt hypotheses were derived from reviewed failures	Freeze a new independently authored holdout before further tuning
Seed/paraphrase dependence	Notebook set contains related variants	Report cluster-level confidence and evaluate semantically independent cases
Model/runtime variability	Historical workbook predictions differ from a fresh original-prompt run	Pin model snapshot, parameters, package versions, and prompt hash
Sparse tail evidence	Only one observed notebook regression	Use confidence intervals and add high-risk slices before release
Offline-to-online gap	No real routing outcomes or transfer costs	Shadow deploy, sample traces, and measure resolution and transfer metrics

Why the excluded evidence is excluded

Later prompt versions were informed by observed benchmark behavior, some clean-room recovery runs were incomplete because LangSmith rejected trace ingestion, and challenge cases were designer-visible. Those artifacts can inform future work, but using them to claim blind generalization or a perfect final score would overstate the evidence. The submission retains only results with defensible provenance and complete paired accounting.

15. Recommended production evaluation plan

1. Publish one adjudicated routing policy with examples for every known cross-category boundary.
2. Freeze the current candidate prompt, model snapshot, parameters, dependencies, and output schema before viewing any new labels.
3. Commission an independently authored holdout with one row per semantic intent rather than paraphrase families, plus deliberate high-risk slices.
4. Use two human reviewers for ambiguous cases and adjudicate disagreements before model scoring.
5. Run the candidate once and retain the result whether it passes or fails. Report paired outcomes, per-class metrics, uncertainty, latency, tokens, and cost.
6. Calibrate any LLM judge against human labels before using it as a release signal; report disagreement slices rather than judge accuracy alone.
7. Shadow deploy before automatic routing. Sample traces, monitor route overrides and transfers, and maintain a human annotation queue.

Suggested online scorecard

Signal	Definition	Suggested action threshold
Human override rate	Share of routed tickets whose category is changed by an agent	Investigate any class with sustained increase
Transfer rate	Tickets transferred after initial routing	Prioritize high-volume or high-severity routes
Time to correct queue	Elapsed time until the ticket reaches the resolving workflow	Alert on tail-latency and category regressions
Resolution / reopen rate	Ticket resolved without reopening	Compare candidate against current routing policy
Invalid or low-confidence output	Schema failure or confidence below routing threshold	Route to human triage; block automation
Cost and latency	Per-ticket token cost and p50/p95 response time	Budget by traffic volume and SLA

16. Final conclusion

The strongest result is not the 98% score by itself. It is the combination of a measurable seven-point gain, full paired outcome accounting, explicit recognition of one regression, class-level diagnosis of the remaining boundary, operational cost analysis, and a release decision that honors a preregistered gate.

The work also exposed a deeper product lesson: evaluation quality depends on policy quality. The same compound request is labeled differently across the workbook and notebook. Until the business decides which workflow owns that request, a model can be penalized for following one reasonable policy and rewarded for another. The next investment should therefore combine policy adjudication with an independent holdout, rather than further tuning on visible development cases.

Final recommendation. Keep focused V2 as a documented candidate. Do not claim hidden-test or production readiness. Resolve the routing contract, freeze the candidate, evaluate once on independent data, and require the regression, class-floor, and operating-cost gates to pass before release.

Artifact and evidence index

Artifact	Purpose	Repository path
Completed workbook	150-case human audit, taxonomy, labels, prompt hypothesis	workbook/Week4_Customer_Support_Evaluation_Kajal_completed.xlsx
Executed notebook	Frozen V1/V2 workflow and saved outputs	notebook/week4_customer_support_evals_executed.ipynb
Paired notebook export	Case-level wins, regressions, stable passes, failures	evidence/results/baseline_vs_improved.csv
Notebook validation summary	Counts, IDs, hashes, and failure records	evidence/results/validation_summary.json
Workbook-aligned comparison	150-case controlled V1/V2 outputs and metrics	evidence/results/official150_case_comparison.csv
LangSmith screenshots	Aggregate, transitions, and trace-level evidence	evidence/screenshots/
Prompt versions	Executed notebook and workbook-focused prompt policies	prompts/
Integrity metadata	Package manifest and SHA-256 records	metadata/

Live links: [Repository](#) | [Google Doc](#) | [LangSmith comparison](#)